Communication

Date	04 July 2026
Team ID	ramanharshitha2005@gmail.com
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Team Standup	Daily	GitHub commit messages + personal task log	Solo Intern (Raman Harshitha)	Track daily progress, update task status, note blockers
2	Progress Update	Weekly	Email / SmartBridge portal	Intern + SmartBridge Mentor	Share weekly milestone completion status and next steps
3	Issue / Bug Discussion	As Needed	GitHub Issues + personal notes	Intern + Mentor (async)	Document and resolve technical blockers (model training errors, API issues, JS bugs)
4	Stakeholder Review	Bi-Weekly	Video call (Google Meet / Zoom)	Intern + SmartBridge Evaluator	Review completed features against project requirements; get feedback
5	Final Demo Rehearsal	Once (end of Week 5)	Local run + browser demo	Solo Intern	Dry-run of the full demo flow: all 5 pages, all 7 APIs, edge cases
6	Final Submission	End of Week 6	GitHub + SmartBridge Portal	Intern + Coordinator	Submit all code, templates, documentation, and demo recording

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	CNN training on local CPU estimated 10+ hours per epoch for the full PlantVillage dataset (87,867 images).	Switched to Google Colab T4 GPU for CNN training. Training completed in ~2 hours. Downloaded model weights (.h5) and committed to models/ folder. Implemented lightweight pixel-feature RF as always-available fallback.
2	Open-Meteo API returned 'Location not found' for some rural Indian city names (e.g., 'Nellore' works, but village names may not).	Added GPS-based geolocation as primary option. City name is a secondary fallback. User can always enter values manually if weather auto-fill fails.

3	Large dataset files (Crop_recommendation.csv and PlantVillage images) exceed GitHub's 100 MB file limit.	CSV is small (~200 KB) and committed directly. PlantVillage images are excluded via .gitignore and referenced via Kaggle link in README.md. Model .joblib files (~50 MB each) are committed; CNN .h5 is optional and downloaded separately.